

Descriptive Statistics

Data exploration phase - or - EDA (Exploratory Data Analysis)

Descriptive stats is the first step to deal with **SAMPLE**. Typical operation that a data analyst, data scientist, business analyst prefer includes:

Normal $\longleftrightarrow$ Skew (check outliers & remove them) $\rightarrow$ • 1.5 IQR rule (Tukey's method)
• Z Score

- ① Check the type of **DISTRIBUTION** for each numerical column.
- ② Identify and deal with **INAPPROPRIATE** data.
- ③ Identify and deal with **OUTLIERS**. (Tukey's Method | 1.5 IQR rule)
- ④ Identify and deal with **CATEGORICAL** data $\rightarrow$ Dummy Variables using OHE
- ⑤ Identify and deal with **MISSING VALUES** $\rightarrow$ Statistical way ✓ (CND - Mean, Dis - Median, Non num - Mode)
Domain way
Hybrid
- ⑥ Identify and deal with **ORDINAL** data
- ⑦ Discovers any

ASSOCIATION
RELATIONSHIP
PATTERN

between two or more variables (columns) in the given dataset.

Hypothesis Testing

① Significance Level (SL) | α (alpha values)

→ % error value, the project can tolerate
(how much amount of errors/inaccuracies the product/project can tolerate).

→ usually SL values is determined in two ways:

Ⓐ Using Statistical Rule | Assumed values | α values

0.05, 0.01, 0.1
5% 1% 10%

Standard α values.

Samples



Statistics

(Approximation)

[These can be errors or inaccuracies in the prediction / analysis]

Ⓑ Data Science method

Practical SL value based on the quality of the dataset using

CROSS-VALIDATION method. — (Machine Learning)

② Confidence Level (CL)

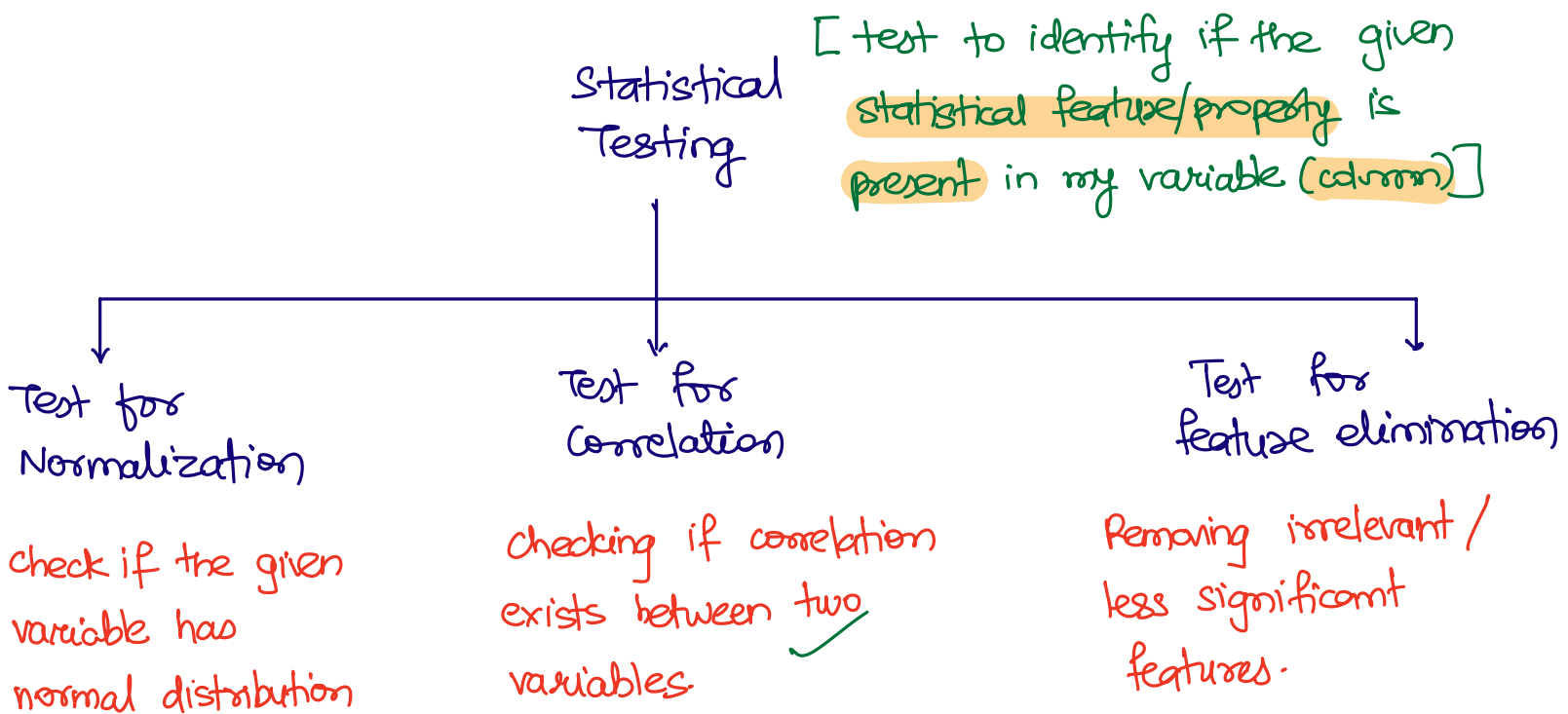
$$CL = 1 - SL$$

[how much minimum accuracy expected from the given data product]

CL determines the minimum evaluation metric threshold to be achieved by the data product.

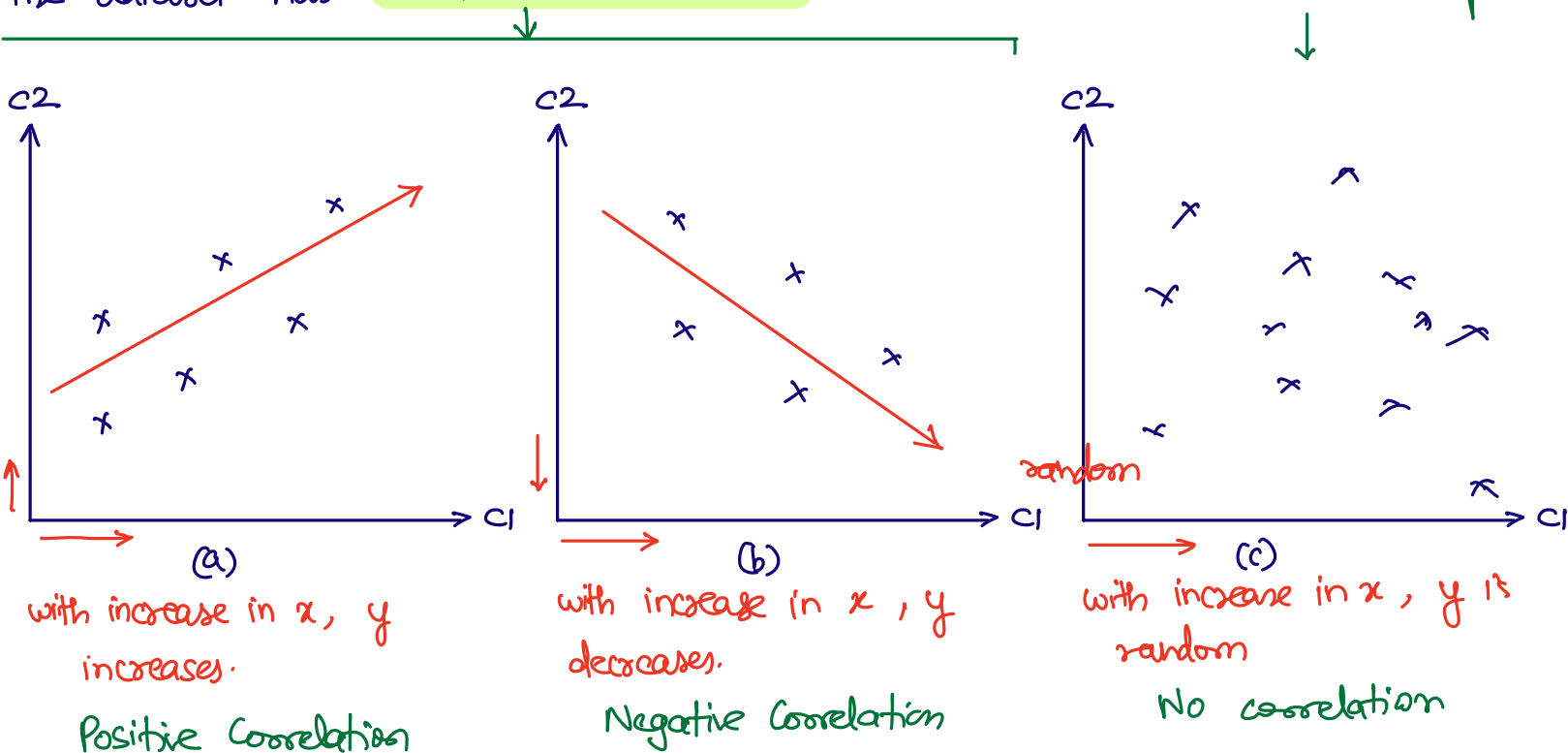
② p-value (probability value)

Calculated SL value derived from statistical testing formula.



Correlation (applicable for Continuous Numerical Data)

Correlation is all about understanding whether the given two variables in the dataset has **LINEAR RELATIONSHIP**.



Goal:
determine whether the evidence
provided by the sample data
variables is STRONG ENOUGH
to support the said assumption.

(assumption)
Hypothesis
Testing

who wins?

NULL
Hypothesis (H_0)



Alternate
Hypothesis (H_a)
(H_1)



* Goal: Test if col1 and col2
have any linear relationship

$H_0 \rightarrow$ Col1 and Col2 have NO linear
relationship.

$H_a \rightarrow$ Col1 and Col2 have linear
relationship



Goal: check if the liquid is an ACID or BASE. }

Note: When it comes to formulating the hypothesis, always remember;

(a) Hypothesis must be formed as BINARY OUTCOME.

(b) Only ONE STATISTICAL PROPERTY must be tested at a time.

Hypothesis 1 $\rightarrow$ check if the liquid is an ACID.
Goal



$H_0 \rightarrow$ Given liquid is NOT an ACID.



$H_a \rightarrow$ Given liquid is an ACID

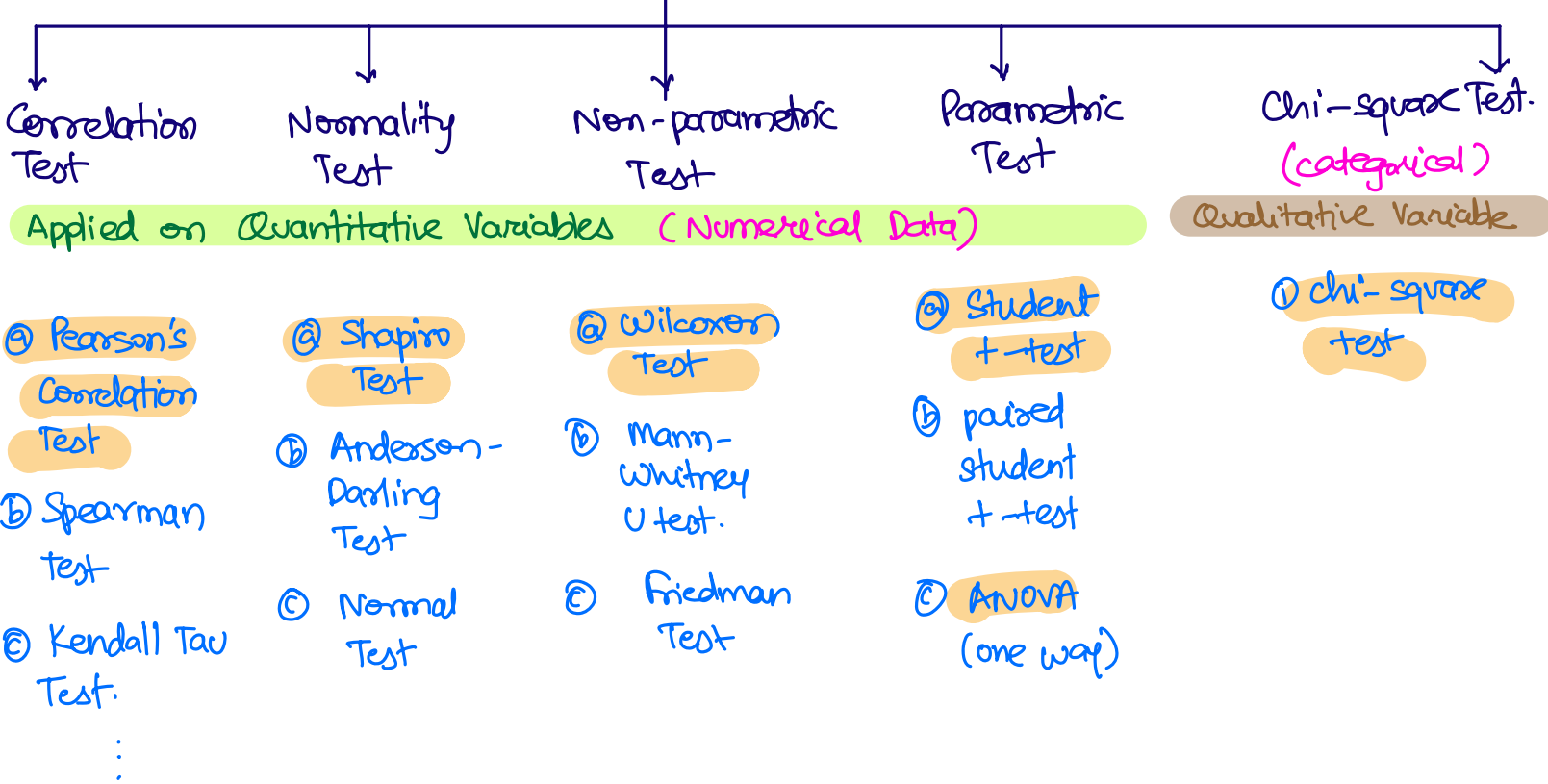
Hypothesis 2 $\rightarrow$ check if the liquid is a BASE
Goal

$H_0 \rightarrow$ Given liquid is NOT a BASE

$H_a \rightarrow$ Given liquid is a BASE

Formulating
the
Hypothesis

Statistical Testing



Goal: Create an AI model that can predict the profit of the company based on company's spending pattern and company's location

